

Penetration of Platinum Complex Anions into Porous Silicon: Anomalous Behavior Caused by Surface-Induced Phase Transition

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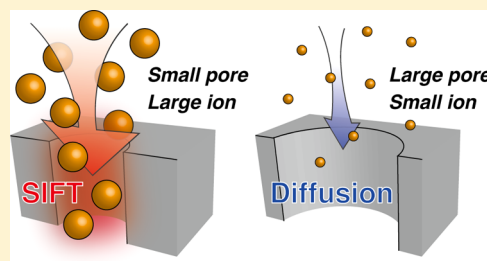
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S Supporting Information

ABSTRACT: We investigate the dynamics of the penetration of platinum complex anions into nanopores during platinum deposition within a nanoporous silicon electrode. The pore-wall surface is hydrophobic and the anions are large enough to behave like hydrophobic solutes. Some of the observations are anomalous in the sense that they cannot be understood in terms of the phenomenological theory for diffusion based on Fick's law. For example, the penetration is faster when the pore diameter is smaller and the anion size is larger. The penetration rate remains unexpectedly fast even when the pores become deeper. The penetration can be made faster using large coexisting cations with sufficiently high hydrophobicity. We show that these results can be interpreted only by statistical mechanics of confined molecular liquids. When the manipulated variables (pore diameter, anion concentration, sizes of platinum complex anions and coexisting cations, etc.) are chosen so that the surface-induced phase transition (SIFT) can take place, the penetration is drastically accelerated. Under the condition with the SIFT occurrence, a strongly attractive, long-ranged effective surface–anion interaction comes into play, leading to the anomalous behavior. The experimental result is in qualitatively good accord with the theoretical argument. The outcome is of vital importance in controlling the mass transfer within nanoporous media and designing next-generation electrochemical devices.



INTRODUCTION

Porous electrodes have played essential roles in electrochemical devices such as batteries, electric double layer capacitors, photocatalysts, and sensors.^{1–7} In maximizing their performance, the control of mass transfer within nanopores is crucially important. For instance, porous carbon electrodes, which are often used in rechargeable batteries, usually exhibit acceptably high electric conductivity. However, the ionic conductivity or mass-transfer rate for ionic species through electrolyte solution within nanopores is very low, so that only a limited proportion of the whole specific surface area contributes to electrochemical reactions.^{8,9} A strategy for enhancing the ionic conductivity, leading to a breakthrough in the design of next-generation electrochemical devices, is highly desired.

Although the specific surface area is an attractive property of porous electrodes, other properties originating from the confinement of electrolyte solutions have started to be studied rather intensively. The behavior of the liquid confined by surfaces is regarded as a key factor to enhance the performances of electrochemical devices using nanoporous electrodes. It is often pointed out that the structure of the liquid near a surface (i.e., the surface-induced structure) is substantially different from that in the bulk. This is evidenced in recent experimental

studies using scanning probe microscopy, nuclear magnetic resonance, ellipsometry, and X-ray reflectometry.^{10–14} Merlet et al. reported that the microstructure of the ionic liquid within a nanopore is different from that at a single surface, which played a pivotal role in enhancing capacitances of nanoporous carbon electrodes.¹⁵ This indicates that the advantageous aspects of porous electrodes cannot be described by a large specific surface area alone. Fedorov and co-workers have argued that the surface-induced structure is substantially influenced by the surface–liquid affinity.^{16–19} For electric double layer capacitors using ionic liquids, Kondrat et al. have reported the possibility of a large increase in the ionic conductivity and enhancing the charging rate in the use of nanoporous electrodes.²⁰ According to their analysis based on molecular dynamics simulations navigated by a phenomenological model, the charging dynamics in nanoporous electrodes is strongly influenced by the ionophobicity of the pore-wall surface. The penetration of ionic liquids into nanopores can be accelerated when the surface is made ionophobic. Such acceleration, which

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could be achieved by adjusting the affinity between the ions and pore-wall surface, is quite attractive from the viewpoint of controlling the mass-transfer rate within nanopores. However, their prediction has not yet been verified by experiments in a comprehensive manner.

We have previously investigated platinum electrodeposition using aqueous electrolyte solution and nanoporous silicon as one of the typical examples of electrochemical processes in nanoporous electrodes and found interesting phenomena that are peculiar to depositions within nanopores: The hydration properties of solutes, corresponding to platinum complex anions in our case, and the affinity of the pore-wall surface with water have essential effects on the deposition behavior.^{21–23} The platinum complex anions are only weakly hydrated due to their large size, and they behave more like hydrophobic solutes and tend to be excluded from the bulk to a hydrophobic surface. Consequently, the concentration of the platinum complex anions is greatly enriched near the pore-wall surface when it is made hydrophobic. As the anion concentration in the bulk increases or the hydrophobicity of the pore-wall surface becomes stronger, the enrichment near the surface is more pronounced. Beyond a threshold anion concentration in the bulk or threshold strength of the hydrophobicity, nanopores are filled with the second phase where the anion concentration is orders of magnitude higher than that in the bulk. This behavior is referred to as the surface-induced phase transition (SIFT).^{24–30} The deposition characteristics below and above the threshold are totally different from each other. In other words, the deposition behavior exhibits a sudden step-change at the threshold. The platinum deposition performed under the condition with the SIFT occurrence should be remarkably accelerated, while without the assistance of the SIFT, the deposition rate is much slower because the supply of anions from the bulk into nanopores through a diffusion based on Fick's law which is a system response to the concentration gradient (this is simply referred to as "diffusion" hereafter) is difficult. This is particularly true for narrow and deep pores. The threshold anion concentration becomes lower as the anion size increases (i.e., anions become more hydrophobic).²² Moreover, the threshold anion concentration can largely be influenced by the coexisting cation species: It becomes lower when the cation species possesses a sufficiently large size and high hydrophobicity.²³ We elucidated these experimental results using a statistical-mechanical theory of confined liquids in which a molecular model is employed for water.

On the other hand, our previous studies^{21–23} for elucidating the SIFT roles in platinum deposition within the nanoporous silicon electrode did not report on its dynamical aspects during the electrochemical reaction. The occurrence of the SIFT was experimentally judged only by the existence of platinum deposits within the whole nanopore space by SEM observations: The process of penetration of platinum complex anions, whose dependence on the pore diameter is an interesting issue, was not investigated. In the present article, we investigate the dynamic aspect of the penetration of platinum complex anions into nanopores during platinum deposition within a nanoporous silicon electrode. The nanopores of the electrode are first filled with ultrapure water, and then the electrode is immersed in aqueous electrolyte solution containing the platinum complex anions for platinum deposition within nanopores. During the penetration of platinum complex anions, we analyze the deposition behavior from various angles. By analyzing time developments of potential, we examine the

effects of the pore diameter, size of the platinum complex anions, concentration of the platinum complex anions in the bulk, coexisting cation species, and pore depth. The results can be understood by the concept of diffusion (a popular phenomenological approach) under the condition without the SIFT occurrence. However, under the condition with the SIFT occurrence, the results can be described only by use of statistical mechanics of confined molecular liquids. Strikingly, the results are often qualitatively opposite to those predicted by the concept of diffusion. For example, the penetration of platinum complex anions into nanopores is faster when the anion size is larger or the pore diameter is smaller. The results from the present study are not directly applicable to the ionic liquid medium due to an absence of distinction between a solute and a solvent. However, they give a novel strategy for designing an electrochemical system using a nanoporous electrode.

The rest of the present article is organized as follows: The concept of the SIFT for a nanopore, which has been revealed by our experiments and analyses based on a statistical-mechanical theory of confined molecular liquids, is explained; dynamic behavior of the penetration process is then predicted; our experimental procedure is described; the experimental results are shown to be in accord with the theoretical prediction; and the significance of the outcome is described with the final discussion and summary.

■ SURFACE-INDUCED PHASE TRANSITION WITHIN A NANOPORE

The SIFT was discussed in detail using the angle-dependent integral equation theory (ADIET)^{31–36} in our earlier publications.^{24–30} The ADIET, in which a molecular model is employed for water, has been quite successful in a variety of studies related to hydrophobic and hydrophilic hydrations. There are three types of SIFT: the partial wetting^{24,25,29,30} and bridging^{26–29,37} transitions and the transition within a nanopore.^{22,23} Here, we present a brief summary focused on the last type. Porous material (e.g., porous silicon) is immersed in aqueous electrolyte solution containing platinum complex anion species with sufficiently large sizes, such as $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$. As the anion size becomes larger with its charge kept constant, hydrogen atoms in a water molecule with positive partial charges can come less close to it, leading to weaker stabilization by the electrostatic attractive potential. That is, a larger anion exhibits weaker affinity with water and becomes less hydrophilic. $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$ possess considerably large sizes and behave more like hydrophobic solutes. The pore-wall surface is hydrophobic. We consider the thermodynamic state of the bulk aqueous electrolyte solution where it is stable as a single phase and subject to no phase transition. Further, it is postulated that no platinum deposition proceeds and the system is in the equilibrium state.

Transition Diameter. When the pore diameter is gradually decreased with the anion concentration in the bulk kept constant (Case A), at a threshold diameter the pore space is abruptly filled with the second phase: The anion concentration is orders of magnitude higher in the second phase than in the bulk. This is the SIFT. We refer to this threshold diameter as the transition diameter.

Transition Concentration. On the other hand, when the anion concentration in the bulk is gradually increased with the pore diameter kept constant (Case B), at a threshold concentration the pore space is abruptly filled with the second

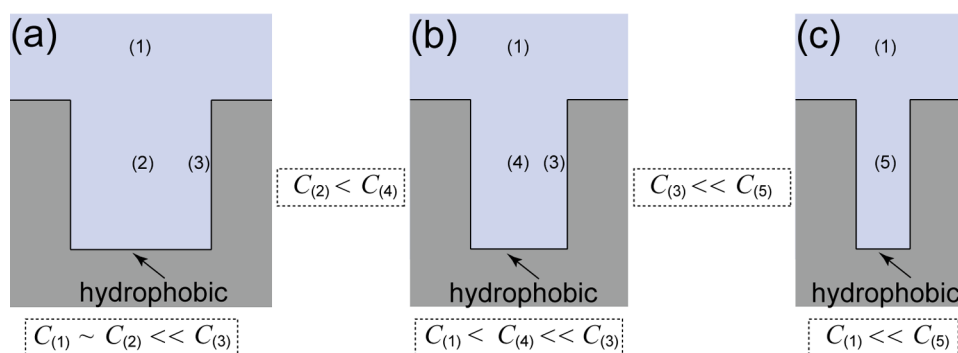


Figure 1. Schematic illustration of anion concentrations at different positions. Nanopores with three different diameters are considered. Anion concentrations in the bulk share the same value. Below a threshold diameter, i.e., (c), the pore space is filled with the second phase.

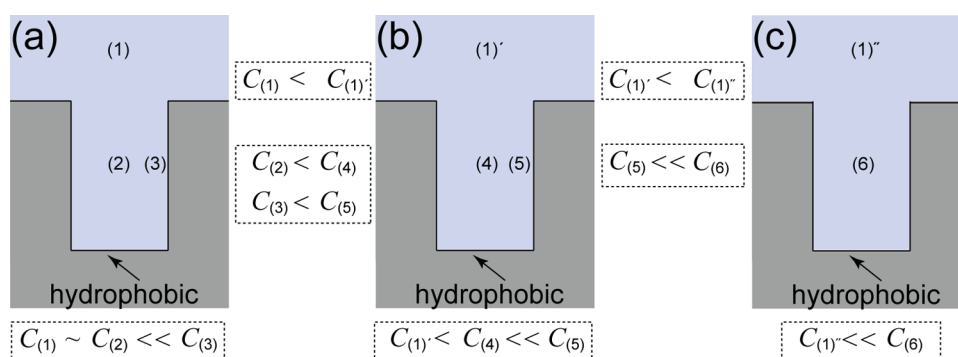


Figure 2. Schematic illustration of anion concentrations at different positions. Three different anion concentrations in the bulk are considered. Pore diameters share the same value. Above a threshold concentration, i.e., (c), the pore space is filled with the second phase.

phase. The threshold concentration is referred to as the transition concentration.

Above the Transition Diameter (Case A). We consider aqueous electrolyte solution containing anions which behave more like hydrophobic solutes. Nanopores with two different diameters larger than the transition diameter are present in the solution. They are illustrated in Figure 1a,b. The pore-wall surface is hydrophobic. The pores in (a) and in (b) possess larger and smaller diameters, respectively. We consider positions (1) through (4): (1) is in the bulk, (3) is in the immediate vicinity of the pore-wall surface, (2) is at the center of the space of the larger pore, and (4) is at the center of the smaller pore. The anion concentrations at these positions follow the orders, $C_{(3)} \gg C_{(2)} \approx C_{(1)}$, $C_{(3)} \gg C_{(4)} > C_{(1)}$, and $C_{(4)} > C_{(2)}$, where $C_{(I)}$ denotes the anion concentration at position (I). Since the anions are only weakly hydrated due to their large sizes, they are excluded from the bulk to the surface: The anion concentration is greatly enriched at the surface.

Below the Transition Diameter (Case A). The average anion concentration within the pore space gradually increases as the pore diameter becomes smaller. However, below a threshold diameter, a drastic change is exhibited: As illustrated in Figure 1c, the pore space is filled with the second phase and $C_{(5)} \gg C_{(3)}$. Position (5) is within the pore space (not necessarily in the immediate vicinity of the pore-wall surface or at the center).

Below the Transition Concentration (Case B). A nanopore with a given diameter is present in the solutions possessing two different anion concentrations in the bulk, which are lower than the transition concentration, illustrated in Figure 2a,b. The anion concentration in the bulk is higher in (b) than in (a). We consider the positions (1), (1)', and (2)

through (5): (1) and (1)' are in the bulk, (3) and (5) are in the immediate vicinity of the pore-wall surface, (2) and (4) are at the center of the pore space. The anion concentrations at these positions follow the orders, $C_{(3)} \gg C_{(2)} \approx C_{(1)}$, $C_{(5)} \gg C_{(4)} > C_{(1)}$, $C_{(4)} > C_{(2)}$, and $C_{(5)} > C_{(3)}$ ($C_{(1)'} > C_{(1)}$).

Above the Transition Concentration (Case B). The average anion concentration within the pore space gradually increases as the anion concentration in the bulk becomes higher. However, above the transition concentration, a drastic change is exhibited: As illustrated in Figure 2c, the pore space is filled with the second phase and $C_{(6)} \gg C_{(5)}$ ($C_{(1)''} > C_{(1)'}$). Position (1)'' is in the bulk and position (6) is within the pore space (not necessarily in the immediate vicinity of the pore-wall surface or at the center).

Effects of Anion Size. As the anion size becomes larger and more hydrophobic, the transition concentration decreases while the transition diameter increases. For example, with the same anion concentration in the bulk, the situation of Figure 1c occurs at a larger pore diameter for $[\text{PtBr}_4]^{2-}$ than for $[\text{PtCl}_4]^{2-}$. In other words, for a certain range of pore diameters, the situation is as in Figure 1c for $[\text{PtBr}_4]^{2-}$, whereas it is as in Figure 1(b) or (c) for $[\text{PtCl}_4]^{2-}$.

Effects of Coexisting Cations. The anion concentration is orders of magnitude higher in the second phase than that in the bulk. In earlier work, we found that not only the anions but also the cations are remarkably enriched in the second phase.²³ Though the charge neutrality is not necessarily satisfied within the confined aqueous solution, the remarkable enrichment of the anions alone would give rise to the energetic instability due to electrostatic repulsive interactions among the anions though they are somewhat screened by water, requiring a sufficiently large amount of cations. The transition diameter and

concentration are not affected by the cation species when the cations are sufficiently small and therefore hydrophilic (e.g., Na^+ and K^+). In the presence of large, rather hydrophobic cations such as $(\text{CH}_3)_4\text{N}^+$ and $(\text{C}_2\text{H}_5)_4\text{N}^+$, the transition diameter is enlarged and the transition concentration becomes lower. The degrees of the enlargement and the lowering are dependent on the cation species. A more hydrophobic cation species elicits a larger degree. (As the cation size becomes larger with its charge kept constant, oxygen atoms in a water molecule with negative partial charges can come less close to it, leading to weaker stabilization by the electrostatic attractive potential. That is, a larger cation exhibits weaker affinity with water and becomes less hydrophilic. $(\text{CH}_3)_4\text{N}^+$ and $(\text{C}_2\text{H}_5)_4\text{N}^+$ behave more like hydrophobic solutes.)

Magnitudes of Transition Diameter and Concentration. Our earlier experimental studies indicated that with a pore diameter of 3 nm the transition concentrations for $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$ (the salts adopted are K_2PtCl_4 and K_2PtBr_4 , respectively) are in the ranges of 0.009–0.010 M and 0.004–0.005 M, respectively.²² The transition concentration is the anion concentration in the bulk at which the SIFT occurs. As the anion concentration in the bulk increases, the transition diameter (i.e., the pore diameter at which the SIFT occurs) becomes larger. An important concern in the present article is to know the magnitude of the transition diameter at a significantly high concentration of $[\text{PtCl}_4]^{2-}$ or $[\text{PtBr}_4]^{2-}$.

Strongly Attractive, Long-Ranged Surface–Anion Effective Interactions. On the basis of the results from theoretical analyses^{26–29,37} on the SIFT and from our recent experimental studies,^{22,23} we discuss the surface–anion effective interaction. When the concentration of $[\text{PtCl}_4]^{2-}$ or $[\text{PtBr}_4]^{2-}$ in the bulk is gradually increased as in Case B, a nanopore is filled with the second phase at the transition concentration. Even if it is further increased, the system can exist as a metastable state without the second phase within the nanopore. A further increase leads to the spinodal concentration beyond which the system can never exist without the second phase. As the system state approaches the spinodal concentration, the surface–anion total correlation function $h_{\text{SA}}(l)$ (l is the distance from an arbitrary point fixed on the pore-wall surface to any point in aqueous electrolyte solution) becomes divergently large (even larger within the nanopore) and infinitely long ranged. The anion concentration divided by that in the bulk $g_{\text{SA}}(l) = h_{\text{SA}}(l) + 1$ exhibits the same behavior. The water-mediated, surface–anion potential of mean force $\phi_{\text{SA}}(l)$ (i.e., an effective surface–anion interaction), which is related to $g_{\text{SA}}(l)$ through

$$g_{\text{SA}}(l) = \exp\{-\phi_{\text{SA}}(l)/(k_{\text{B}}T)\} \quad (1)$$

also exhibits the following behavior (k_{B} is the Boltzmann constant and T the absolute temperature): $\phi_{\text{SA}}(l)$ is attractive, divergently strong (even stronger within the nanopore), and infinitely long ranged. This feature of $\phi_{\text{SA}}(l)$ should hold true above the spinodal concentration.

It is worthwhile to emphasize the difference between $\phi_{\text{SA}}(l)$ and the surface–solute direct interaction in vacuum. $\phi_{\text{SA}}(l)$ represents “free energy of aqueous solution for a finite value of l ” minus “that for infinitely large l ”. It is remarkably dependent not only on the surface and anion properties but also on the thermodynamic state of aqueous solution including the anion concentration in the bulk. Near and above the spinodal concentration the range of $\phi_{\text{SA}}(l)$ becomes infinitely long ranged (macroscopic). When the SIFT occurs but the anion

concentration is below the spinodal one, it is difficult to estimate the range of $\phi_{\text{SA}}(l)$. However, it is considerably stronger and longer ranged than the surface–anion direct interaction in vacuum, as described in the next paragraph in more detail. Here, we give two more remarks: The mobility of anions within a nanopore is not decelerated despite the very high anion concentration, which is supported by our experimental observations presented in the later sections; and within a nanopore with hydrophobic surface, the anion must be sufficiently hydrophobic for the SIFT to occur, i.e., its solubility must be sufficiently low. For example, the SIFT will never occur for F^- or Cl^- .

In the real system, the SIFT unavoidably occurs at the transition concentration which is lower than the spinodal one. Even at the transition concentration, the attractive, effective interaction $\phi_{\text{SA}}(l)$ is significantly strong and long ranged. Within the nanopore, it is even stronger. Due to this effective interaction, which overcomes the thermal motion of anions disturbing the spatial distribution of anion concentration, the second phase shown in Figure 1c or 2c is retained in the equilibrium state. An important point which becomes useful in the Results and Discussion section is the following: When the concentration of $[\text{PtCl}_4]^{2-}$ or $[\text{PtBr}_4]^{2-}$ in the bulk is gradually increased above the transition concentration, the anion concentration in the second phase still increases, $\phi_{\text{SA}}(l)$ becomes further stronger and longer ranged, and $\phi_{\text{SA}}(l)$ within the nanopore becomes even stronger. In other words, as the concentration exceeds the transition one to a larger extent, $\phi_{\text{SA}}(l)$ including that within the nanopore becomes stronger.

Taken together, well below the transition concentration the interaction range is as short as a few times of the molecular diameter of water, and near and above the spinodal concentration it is infinitely long (macroscopic). Above the transition concentration and well below the spinodal one it is difficult to estimate the range, but should not be macroscopic but be much longer than a few times of the molecular diameter of water. We note that the SIFT occurrence within the ~15 nm pore is experimentally confirmed for $[\text{PtBr}_4]^{2-}$ as shown in the Effect of Cation Species Coexisting with Platinum Complex Anions section. This fact is suggestive that the range can be roughly 15 nm (or even longer) under the experimental condition because such an interaction works only when the SIFT occurs within the nanopores.

■ PENETRATION OF ANIONS INTO NANOPORE

The present experiment is performed in the following way (see the Experimental Section): Pores of porous silicon are hydrophobic because hydrogen atoms terminate silicon at the pore-wall surface. They are first filled with ultrapure water, and then the porous silicon is immersed in electrolyte solution containing the platinum complex anions. Three pore diameters, 4, 15, and 30 nm, are tested. Therefore, penetration of the anions into pores occurs and the anions are consumed on the silicon surface by the platinum deposition. The silicon surface consists of the top surface (i.e., the surface outside the pores) and the pore-wall surface. The platinum deposition proceeds on the top surface as soon as the porous silicon is immersed in the electrolyte solution, whereas it proceeds on the pore-wall surface only after the anions arrive at this surface. Unlike in the description in the Surface-Induced Phase Transition within a Nanopore section, the system is not in an equilibrium state: Penetration of anions occurs and the platinum deposition proceeds. That is, dynamics is observed in the present

experiment. Nevertheless, the discussion based on statistical mechanics of liquids is very useful as explained below.

Description as Diffusion Process. A diffusion equation based on Fick's law has widely been used for describing the anion penetration into pores. In this description, the concentration gradient of anions is the key quantity acting as the driving force of the penetration. If no platinum deposition proceeds, this description erroneously predicts that in the equilibrium state the anion concentration is uniform, namely, it takes the same value for all the positions considered in Figures 1 and 2.

Description Based on Statistical Mechanics. The penetration could be described if the concentration gradient is replaced by the chemical-potential gradient. However, the coefficient multiplied to the chemical-potential gradient becomes orders of magnitude larger when the SIFT occurs, whose meaning is not physically sound. Therefore, as discussed in "Strong, Long-ranged Effective Surface-anion Attractive Interactions", there are essential factors which can be taken into account only by statistical mechanics of confined liquids. Here, we consider Case B. When the anion concentration in the bulk exceeds the transition concentration, the strong, long-ranged effective surface-anion attractive interaction comes into play toward the formation of the second phase within a nanopore: Within the nanopore, the effective interaction is even stronger. Here, we emphasize that "surface" denotes not a single surface coexisting with the nanopores in the system but the surface within a nanopore (i.e., the pore-wall surface) and "anion" denotes an anion in the system (not necessarily within a nanopore).²⁹ As the anion concentration in the bulk increases (i.e., as it further exceeds the transition concentration or even the spinodal one), the interaction becomes further stronger and longer ranged and the penetration of anions into a nanopore is remarkably accelerated. Even below the transition concentration, the effective surface-anion attractive interaction does exist, but it is much weaker and shorter-ranged.

Let us consider two systems where $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$ are employed, respectively. The concentrations of these anions share the same value and the diameters of the nanopores are also the same. The transition concentration for $[\text{PtBr}_4]^{2-}$ is considerably lower than that of $[\text{PtCl}_4]^{2-}$ because the former is more hydrophobic (according to our estimation,^{21,22} the sizes of $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$ are approximately 0.6 and 0.7 nm, respectively). There are three cases: (I) The anion concentration is below the transition concentration for both of $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$; (II) it is below the transition one for $[\text{PtCl}_4]^{2-}$ but above the transition one for $[\text{PtBr}_4]^{2-}$; and (III) it is above the transition one for both $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$. In Case (I), the effective surface-anion attractive interaction is rather weak and short-ranged, and the anion penetration into nanopores may be argued as the diffusion process. The penetration rate for $[\text{PtBr}_4]^{2-}$ should be similar to or slightly slower than that for $[\text{PtCl}_4]^{2-}$ since the diffusivity scales in inverse proportion to the radius of the diffusing species. This holds true in Cases (II) and (III) as well as when the diffusion equation is applied. However, in these cases applying statistical mechanics of liquids gives the opposite result. In Case (II), the effective surface-anion attractive interaction for $[\text{PtCl}_4]^{2-}$ is rather weak and short-ranged whereas that for $[\text{PtBr}_4]^{2-}$ is strong and long ranged. In Case (III), the effective surface-anion attractive interaction for $[\text{PtCl}_4]^{2-}$ is strong and long ranged, but that for $[\text{PtBr}_4]^{2-}$ is even stronger and longer

ranged. Consequently, penetration of $[\text{PtBr}_4]^{2-}$ into nanopores occur much faster than that of $[\text{PtCl}_4]^{2-}$.

Suppose that $[\text{PtCl}_4]^{2-}$ is employed and the pore diameter is gradually decreased with the anion concentration kept constant. The diffusion equation predicts that the anion penetration into pores will be progressively decelerated as the diameter becomes smaller. Statistical mechanics of confined liquids, however, gives a substantially different prediction: The anion penetration into pores is decelerated as the diameter becomes smaller, but below the transition diameter it abruptly becomes much faster and accelerated as the diameter reduces further. This is because the effective surface-anion attractive interaction abruptly becomes strong and long ranged below the transition diameter and they become even stronger and longer ranged with the decrease in the diameter.

Strikingly, in the experiment described below, we demonstrate that the above arguments hold true.

■ EXPERIMENTAL SECTION

A porous silicon electrode was prepared by anodization of p-type silicon (100) in 22 wt % HF solution (48 wt % HF: ethanol = 1:1.7 in volume) under constant current density. Three types of porous silicon with different pore sizes were prepared by changing a resistivity of silicon wafer and an applied current density. The average diameters of the pores were ~4, ~15, and ~30 nm when the resistivities of the silicon wafers were 1–4.5, 0.005–0.03, and 0.0015–0.0030 Ω cm, and the current densities were 2.0, 80, and 80 mA cm^{-2} , respectively. The duration of anodization was tuned so that the thickness of porous layers could be adjusted to 1.2 μm . Especially for the case where thicker porous layers with pore sizes of ~4 nm were formed, the porous silicon electrode was prepared by anodization of p-type silicon (100) with resistivity of 1–4.5 Ω cm in 30 wt % HF solution (48 wt % HF: water: methanol = 11:3:6 in volume) at a current density of 14 mA cm^{-2} . The anodized area was 0.79 cm^2 .

As-anodized porous silicon was soaked in ultrapure water for 10 min to extract the residual fluorides. The pores were kept wet during washing because the electrolyte solution for platinum deposition could not infiltrate the pores if they dried up. After washing the porous silicon, displacement deposition of platinum was performed under dark conditions using the porous silicon as the working electrode. The reference electrode was Ag/AgCl sat. KCl. Aqueous solution containing 0.1 M K_2PtCl_4 and 0.5 M NaCl was used for the deposition bath. Aqueous solutions of 0.02 or 0.1 M K_2PtBr_4 and 0.5 M KBr were also used to examine the anion-size effect. Thus, we employed two platinum complex anions, $[\text{PtCl}_4]^{2-}$ and $[\text{PtBr}_4]^{2-}$, as the platinum source. Time development of the potential was measured for the porous silicon working electrode as well as for an unporosified silicon wafer with a potentiostat (Hokuto Denko, HZ-3000).

The amount of platinum deposited within the porous silicon was measured by an electronic balance (Mettler Toledo, XP20UV). The cross-sectional views of porous silicon after displacement deposition of platinum were observed with a field emission-type scanning electron microscope (JEOL, JSM-6500F).

■ RESULTS AND DISCUSSION

Time Development of Open Circuit Potential during Displacement Deposition. Because silicon works as a weak

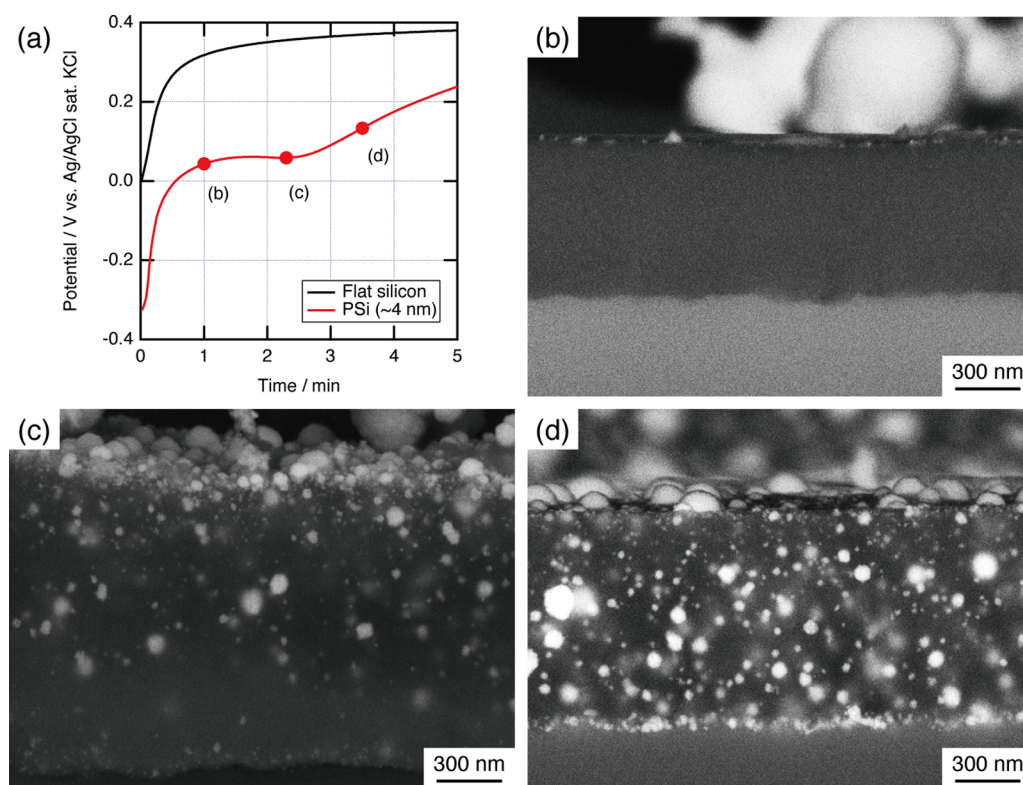
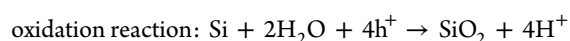
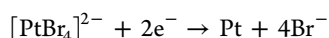
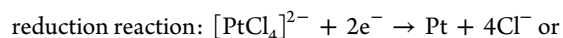


Figure 3. (a) Time developments of the mixed potential measured for silicon wafer (flat silicon) and porous silicon (PSi) whose average pore size is ~ 4 nm. (b–d) Cross-sectional compositional images of platinum in stages (b–d) indicated by the red closed circles in (a), respectively. The measurements were carried out in the aqueous solution of “0.1 M K_2PtCl_4 + 0.5 M NaCl”.

reductant, displacement deposition of platinum spontaneously proceeds when the silicon wafer is immersed in the electrolyte solution containing $[\text{PtCl}_4]^{2-}$ or $[\text{PtBr}_4]^{2-}$.³⁸ The reactions of displacement deposition can be written as



When displacement deposition occurs, the reactions proceed at the so-called mixed potential (this is simply referred to as the potential) where the net current is balanced. Figure 3a illustrates the time development of the potential during displacement deposition of platinum using 0.1 M $[\text{PtCl}_4]^{2-}$ as the platinum source. The black and red lines indicate the results for flat silicon and for porous silicon whose average pore size is ~ 4 nm, respectively. Immediately after the deposition starts, an upward shift in potential is commonly observed for both substrates. This potential shift occurs for the following reason: As displacement deposition proceeds, the effective surface area of bare silicon becomes smaller due to the coverage by silicon oxide produced as well as by the platinum deposit. This leads to a decrease in the current density of silicon oxidation, since it is considered that not the silicon oxidation reaction but the platinum deposition reaction is rate limiting (the reason why platinum deposition is rate limiting is discussed in the [Effect of Size of Platinum Complex Anions](#) section). To balance with this decrease, the potential exhibits an upward shift and the reduction current of platinum undergoes a decrease. When bare silicon is no longer available, displacement deposition ceases and the potential stays at a constant value where the

equilibrium between Pt and $[\text{PtCl}_4]^{2-}$ is achieved. We could begin the potential measurement only after a few seconds had passed since the start of displacement deposition in each experimental run. Due to the time lag thus caused, the initial potential varied from run to run. We note, however, that our discussions are not affected by this variation.

A plateau of potential, where the potential stays at a constant value, is observed before displacement deposition ceases only when using the porous silicon electrode. As mentioned in the [Experimental](#) section, the electrolyte solution for platinum deposition was put into the cell after washing the porous silicon electrode with ultrapure water. Therefore, the pores were initially filled with ultrapure water. Unlike in the case of the flat silicon electrode, penetration of the platinum complex anions into the pores is necessitated. We investigate the behavior of platinum deposition in various stages during the deposition because the emergence of the potential plateau seems to be closely related to the porous structure of the electrode. Figure 3b–d depict compositional images of cross sections of the porous silicon electrode prepared at three different stages of displacement deposition. The contrast in the image reflects the density of platinum deposited within the porous layer, so that the brighter spot can be larger than the pore size. Before the potential reaches a plateau (Figure 3b), platinum is deposited only on the top surface of porous silicon. In the final stage of the potential plateau (Figure 3c), platinum deposit is observed within the porous layer as well. After the potential plateau (Figure 3c), the potential shifts upward again. During this upward shift, the density of the platinum deposit within the porous layer exhibits a drastic increase. Judging from the deposition behavior observed in Figure 3b,c, we can conclude

that the potential plateau is induced by the process during which the platinum complex anions penetrate into the pores.

We summarize below the results for the porous silicon electrode (see Figure 3). In the initial stage, the potential remains negative, implying fast platinum deposition. The platinum complex anions are available only on the top surface of the porous electrode because they have not yet penetrated into nanopores, leading to consumption of most of the platinum complex anions on the top surface. For this reason, the upward shift of potential in the initial stage looks similar to that for the flat silicon electrode. During the initial stage, the effective surface area of bare silicon becomes smaller, resulting in a decrease in the net current density of silicon oxidation. To balance with this decrease, the potential exhibits an upward shift and the reduction current of platinum undergoes a decrease, leading to a slower deposition reaction. As a result, the platinum complex anions start to penetrate into the porous layer without being significantly consumed at the pore opening. During the time when the pores are filled with pure water, the oxidation of the silicon wall is not appreciable. However, the oxidation starts to occur when the platinum complex anions (oxidizing agents) become available near the pore-wall surface by penetration. This leads to an increase in the net current for silicon oxidation. The contribution of silicon oxidation from the upper part of the porous layer down to the bottom during the penetration temporarily makes the potential change significantly slower, resulting in the potential plateau. Because the time course of the potential is strongly correlated with the penetration of platinum complex anions, the potential plateau is not stable. After the potential plateau, the concentration of $[\text{PtCl}_4]^{2-}$ within the pores becomes sufficiently high, so the potential starts to shift upward again and finally reaches the equilibrium value. Thus, the duration of the potential plateau includes information on the dynamics of the penetration of platinum complex anions into the pores. Longer duration of the plateau implies slower penetration. Hereafter, we discuss the effects of the pore diameter, size and concentration of the platinum complex anions, coexisting cation species, and pore depth on the penetration dynamics.

Effect of Pore Diameter. Figure 4 shows the time developments of the potential during displacement deposition of platinum in the bath containing $[\text{PtCl}_4]^{2-}$ at 0.1 M as the platinum precursor. Flat and porous silicon electrodes are considered in Figure 4a, and the average pore diameters of the porous silicon electrodes are 4, 15, and 30 nm. For the flat silicon electrode, the potential exhibits a monotonic increase immediately after the deposition is started. In about 2 min, the increase is significantly decelerated and the potential displays a tendency to converge toward ~ 0.4 V that corresponds to the equilibrium potential for the reduction and oxidation of platinum. For the porous silicon electrode whose average pore diameter is 30 nm, the potential plateau is observed in the range from 2 to 7 min after the deposition is started. The plateau is followed by an upward shift to reach the equilibrium potential. When the average pore diameter is decreased to 15 nm, the duration of the potential plateau becomes longer: It is observed in the range from 2 to 13 min. This change can readily be explained because the smaller pores are less advantageous for the diffusive penetration of platinum complex anions into the pores. By contrast, a further decrease in the average pore diameter to 4 nm leads to an unanticipated change in the duration of the potential plateau: It is remarkably shortened (much shorter even than in the case of 30 nm). The

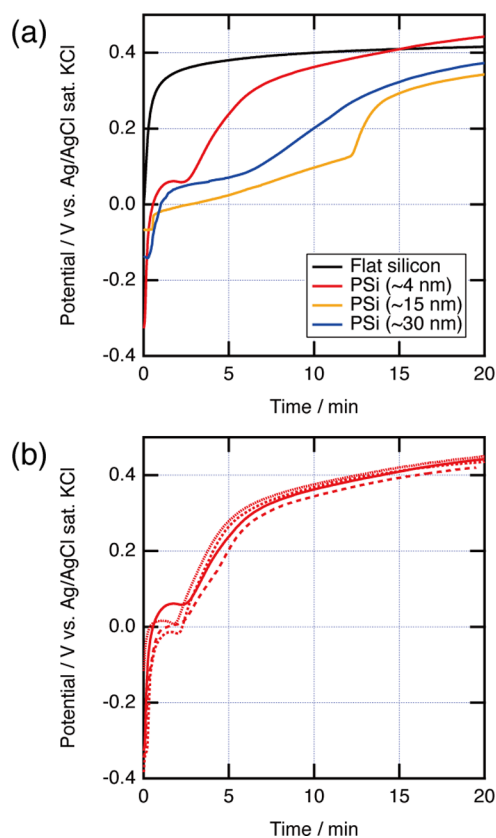


Figure 4. (a) Time developments of the mixed potential measured for silicon wafer (flat silicon) and porous silicon (PSi) whose average pore sizes are ~ 4 , ~ 15 , and ~ 30 nm. (b) Reproducibility of the time development verified for porous silicon with a pore size of ~ 4 nm. The measurements were performed four times. The four curves thus obtained are compared: They are similar to one another even in a quantitative sense. The red solid line is identical with the one in (a). All of the measurements were carried out in aqueous solution of 0.1 M K_2PtCl_4 + 0.5 M NaCl.

reproducibility of the time development in the case of 4 nm was verified by repeated measurements as illustrated in Figure 4b. While the variation in pore size is achieved using Si substrates with different resistivities, the Ohmic potential drop within Si which is included in the determination of mixed potential can be neglected, because the anodic and cathodic sites are very closely located.³⁷ Therefore, these results indicate that quite small pores of silicon are the most advantageous for the penetration of platinum complex anions.

When the average pore diameter is 30 or 15 nm, the penetration dynamics for the platinum complex anions can qualitatively be discussed within the framework of diffusion. However, this is not true in the case of 4 nm pores. For the aqueous solution containing $[\text{PtCl}_4]^{2-}$ at 0.1 M, the average pore diameter of 4 nm is small enough such that the SIFT takes place. As explained in the Strongly Attractive, Long-Ranged Surface–Anion Effective Interactions section, the surface-anion potential of mean force (i.e., effective interaction) $\phi_{\text{SA}}(l)$ becomes strongly attractive and long ranged, and the penetration of platinum complex anions into the pores is drastically accelerated.

Effect of Size of Platinum Complex Anions. Figure 5a shows the time developments of the potential during displacement deposition of platinum in the baths containing $[\text{PtCl}_4]^{2-}$ at 0.1 M and $[\text{PtBr}_4]^{2-}$ at 0.1 M. Porous silicon

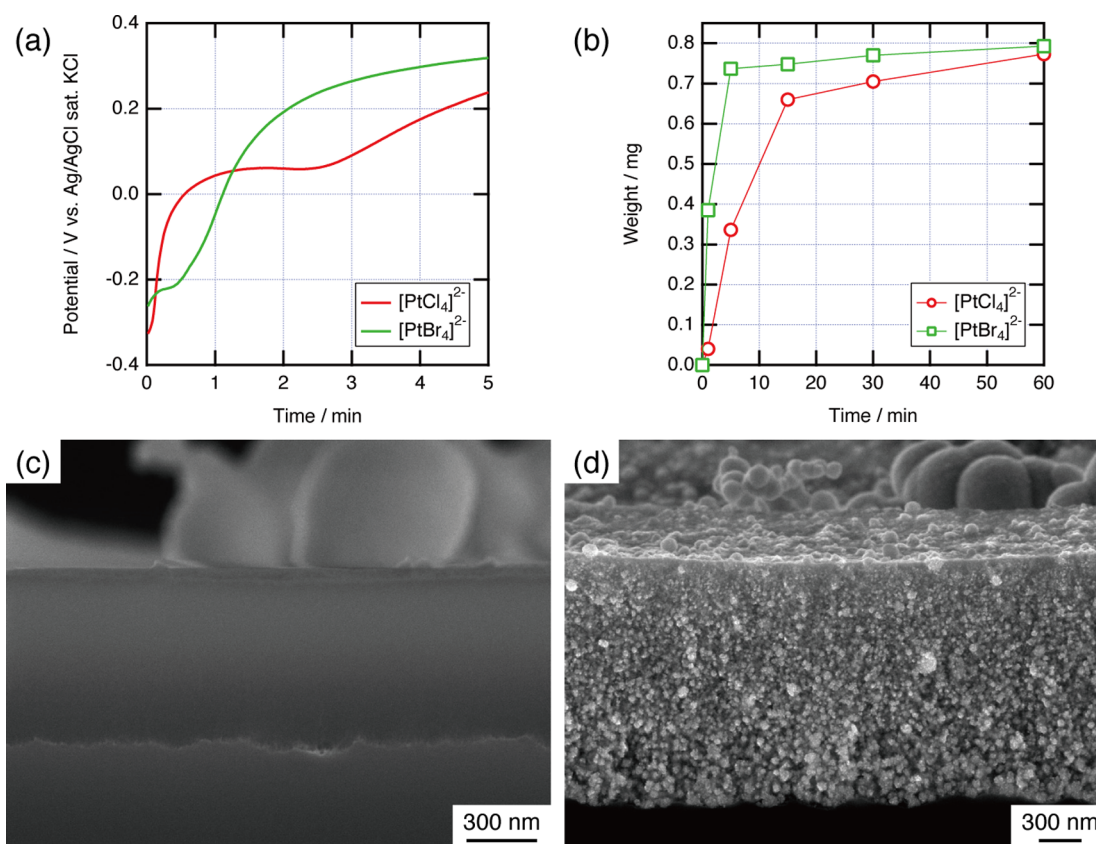


Figure 5. Results for porous silicon whose average pore size is ~ 4 nm. (a) Time developments of the mixed potential measured. The measurements were carried out in aqueous solutions of 0.1 M K_2PtCl_4 + 0.5 M NaCl ($[\text{PtCl}_4]^{2-}$) and 0.1 M K_2PtBr_4 + 0.5 M KBr ($[\text{PtBr}_4]^{2-}$). (b) Amount of platinum deposited on porous silicon plotted as a function of the time. (c and d) Secondary electron images of cross-sectional view of porous silicon after 1 min for (c) $[\text{PtCl}_4]^{2-}$ and for (d) $[\text{PtBr}_4]^{2-}$.

electrodes are used, and the average pore diameter is 4 nm. The size of $[\text{PtBr}_4]^{2-}$ is slightly larger than that of $[\text{PtCl}_4]^{2-}$. The duration of the potential plateau observed for $[\text{PtBr}_4]^{2-}$ is only ~ 10 s which is roughly an order of magnitude shorter than that observed for $[\text{PtCl}_4]^{2-}$. The time variations of the amount of platinum deposit are plotted in Figure 5b. In the case of $[\text{PtBr}_4]^{2-}$, the amount abruptly increases within 1 min and almost reaches its maximum value in about 5 min. The increase is much slower in the case of $[\text{PtCl}_4]^{2-}$. In both cases, platinum is uniformly deposited within the porous layer without plugging of nanopores by deposits at the pore openings, and the surface areas available to the oxidation reaction share almost the same value. Therefore, the amounts of platinum deposit converge to the same value. This means that the shorter duration of the potential plateau for $[\text{PtBr}_4]^{2-}$ is attributed not to the plugging but to the fast penetration of platinum complex anions. Figure 5c,d compares cross-sectional views of the samples taken after 1 min has passed. In the case of $[\text{PtCl}_4]^{2-}$, the potential plateau has just started and no platinum deposit is observed within the porous layer. In the case of $[\text{PtBr}_4]^{2-}$, the plateau has already been finished and the pores are almost completely filled with platinum. Note that this result indicates that platinum deposition reaction is rate limiting. This is because the time required for the amount of platinum deposits to reach its maximum would be the same regardless of the platinum complex species if the Si oxidation reaction was rate limiting.

The experimental results described above indicate that the larger anions ($[\text{PtBr}_4]^{2-}$) penetrate into the pores more rapidly than the smaller anions ($[\text{PtCl}_4]^{2-}$). This indication is

counterintuitive, because the opposite should be true at least according to the Stokes equation. It can be understood only in terms of the SIFT. With the average pore diameter of 4 nm and the anion concentration of 0.1 M, the SIFT occurs in both cases. However, the transition concentration is considerably lower, or equivalently, the anion concentration exceeds the transition one to a larger extent, for $[\text{PtBr}_4]^{2-}$. It follows from the argument in the Strongly Attractive, Long-Ranged Surface–Anion Effective Interactions section that the surface–anion effective interaction $\phi_{\text{SA}}(l)$, which is attractive, becomes stronger and longer ranged and the penetration of platinum complex anions into the pores is more accelerated in the case of $[\text{PtBr}_4]^{2-}$.

Effect of Concentration of Platinum Complex Anions.

We then look at the effect of the anion concentration in the bulk. Figure 6a shows the time developments of the potential during displacement deposition of platinum in baths containing $[\text{PtCl}_4]^{2-}$ at 0.04, 0.06, and 0.1 M. Figure 6b shows those in the baths containing $[\text{PtBr}_4]^{2-}$ at 0.04, 0.06, and 0.1 M. Porous silicon electrodes are used and the average pore diameter is 4 nm. The SIFT occurs in all cases. The duration of the potential plateau always becomes shorter with an increase in the concentration of platinum complex anions. For a given concentration, the larger anions ($[\text{PtBr}_4]^{2-}$) penetrate into the pores more rapidly because the duration of the plateau is shorter. These results indicate that as the anion concentration exceeds the transition concentration to a larger extent, the attractive $\phi_{\text{SA}}(l)$ becomes stronger and longer ranged. It is interesting to note the following: The potential plateau

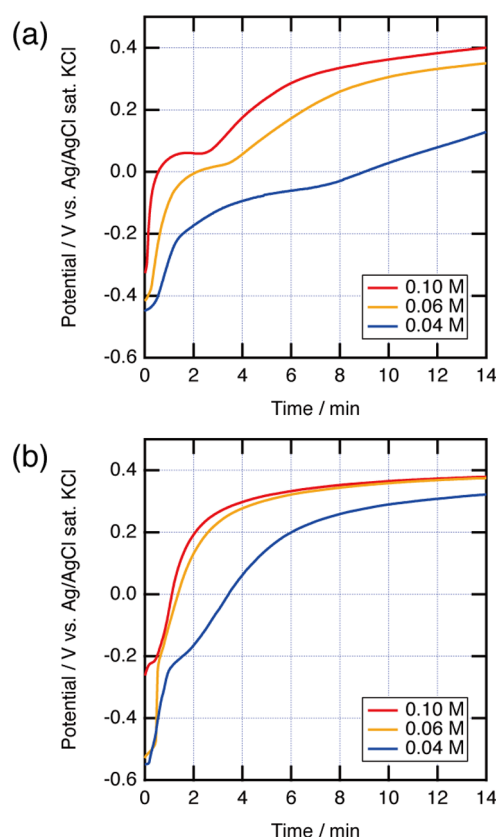


Figure 6. Time developments of the mixed potential measured for porous silicon whose average pore size is ~ 4 nm. The measurements were carried out in the aqueous solutions of (a) X M K_2PtCl_4 + 0.5 M NaCl and (b) X M K_2PtBr_4 + 0.5 M KBr. X is 0.10 , 0.06 , or 0.04 .

observed under the condition, 0.04 M $[\text{PtCl}_4]^{2-}$ and 4 nm (blue curve in Figure 6a), is much shorter than that under the condition, 0.1 M $[\text{PtCl}_4]^{2-}$ and 15 nm (yellow curve in Figure 4a). This result cannot be interpreted by the concept of diffusion, because the diffusion of platinum complex anions should be much faster under the latter condition due to the higher concentration gradient and larger pores.

Figure 7 compares time developments of the potential during displacement deposition of platinum under the two conditions, 0.04 M $[\text{PtCl}_4]^{2-}$ and 4 nm, and 0.04 M $[\text{PtBr}_4]^{2-}$ and 15 nm. The SIFT certainly occurs under the former condition as discussed above. The potential plateau is observed in the range from 2 to 8 min and its duration is only 6 min. Under the latter condition, it is observed in the range from 2 to 5 min: The duration of the plateau is even shorter (3 min). Hence, the SIFT occurs under the latter condition as well. It follows that the transition pore diameter can be as large as 15 nm, much larger than the nanometer scale in the case of very hydrophobic anion species.

Effect of Cation Species Coexisting with Platinum Complex Anions. In our earlier work, it was shown that the threshold concentration for the SIFT is substantially influenced by the cation species coexisting with the platinum complex anions. It becomes lower as the cation size increases. Displacement deposition of platinum was carried out in electrolyte solution containing 0.04 M of $[\text{PtCl}_4]^{2-}$ and 0.5 M of a coexisting cation species. Porous silicon electrodes were used and the average pore diameter was 4 nm. Figure 8 compares the time developments of the potential during

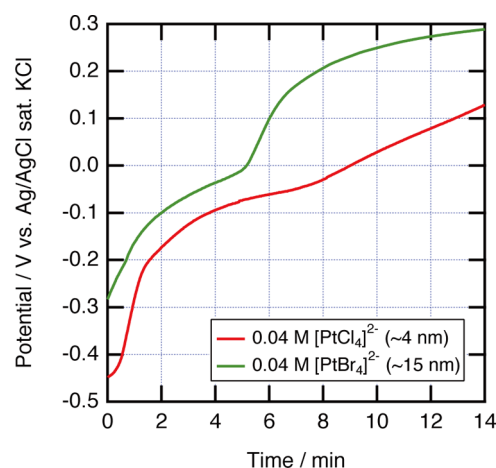


Figure 7. Time developments of the mixed potential measured for porous silicon in aqueous solution. Red curve: The average pore size is ~ 4 nm and the aqueous solution is 0.04 M K_2PtCl_4 + 0.5 M NaCl. Green curve: The average pore size is ~ 15 nm and the aqueous solution is 0.04 M K_2PtBr_4 + 0.5 M KBr.

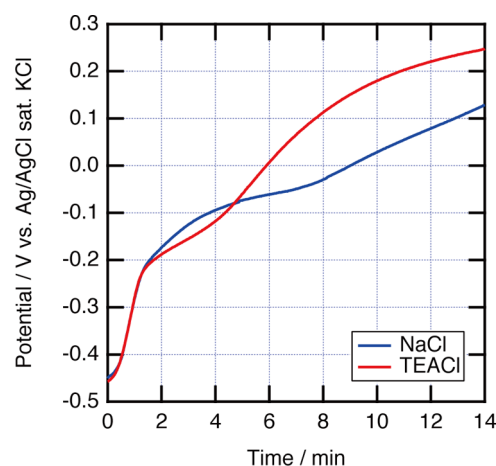


Figure 8. Time developments of the mixed potential measured for porous silicon whose average pore size is ~ 4 nm. The measurements were carried out in the aqueous solutions of 0.04 M K_2PtCl_4 + 0.5 M NaCl and 0.04 M K_2PtCl_4 + 0.5 M TEACl.

displacement deposition of platinum for which Na^+ and $(\text{C}_2\text{H}_5)_4\text{N}^+$ are chosen as the cation species. The penetration of platinum complex anions into the pores is significantly accelerated when Na^+ is replaced by $(\text{C}_2\text{H}_5)_4\text{N}^+$. The potential plateau lasts until 8 min for Na^+ whereas it only lasts until 4 min for $(\text{C}_2\text{H}_5)_4\text{N}^+$. The transition concentration is considerably lower for $(\text{C}_2\text{H}_5)_4\text{N}^+$: The anion concentration exceeds the transition one to a larger extent. Consequently, the attractive surface–anion effective interaction $\phi_{\text{SA}}(l)$ becomes stronger and longer ranged and the penetration of platinum complex anions into the pores is accelerated more in the case of $(\text{C}_2\text{H}_5)_4\text{N}^+$.

Effect of Pore Depth. We then discuss the effect of the pore depth by changing the thickness of porous silicon. Even when the thickness is largely changed, the pore size remains constant from the top to the bottom. Therefore, the increase in the thickness is proportional to that in the pore depth. We prepared porous silicon whose thickness and average pore size were ~ 14 μm and 4 nm, respectively. This new porous silicon is over 10 times thicker than the one used in the preceding

sections. According to the concept of diffusion, when the pore becomes over 10 times deeper, the potential plateau could last at least 100 times longer because the diffusion time grows as square of the pore length. Figure 9a compares time develop-

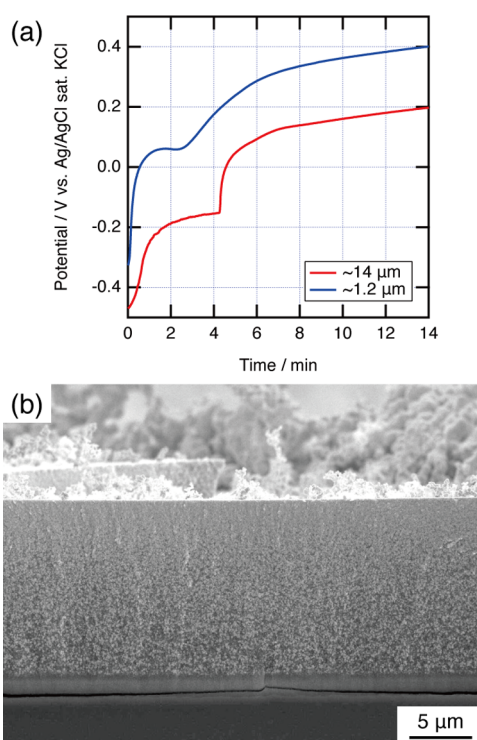


Figure 9. Time developments of the mixed potential measured for porous silicon whose average pore size is ~ 4 nm. Average pore depths of the porous silicon are $\sim 14 \mu\text{m}$ (red curve) and $\sim 1.2 \mu\text{m}$ (blue curve). The measurements were carried out in aqueous solution of $0.1 \text{ M K}_2\text{PtCl}_4 + 0.5 \text{ M NaCl}$. (b) Secondary electron image of cross-sectional view of porous silicon whose average pore size and depth are ~ 4 nm and $\sim 14 \mu\text{m}$, respectively, after 60 min of platinum deposition.

ments of the potential during displacement deposition of platinum for the thinner ($\sim 1.2 \mu\text{m}$) and thicker ($\sim 14 \mu\text{m}$) porous silicon substrates. Surprisingly, the duration of the plateau is about 3 min which is only twice as long than in the case of thinner porous silicon though the platinum deposits are observed from the opening to the bottom of the nanopores as shown in Figure 9b. This result gives another evidence of the presence of strongly attractive, long-ranged surface-anion effective interaction.

CONCLUDING REMARKS

The present study was made possible by the integration of electrochemistry and statistical mechanics of confined molecular liquids and by the comparison between theoretical predictions and experimental results. Liquid confined on the nanometer scale behaves quite differently from the bulk liquid. Solvation properties of solutes in the confined solvent are substantially different from those in the bulk solvent. The emphasis is on dynamics of the penetration of platinum complex anions into a porous silicon electrode. An important point is the adoption of a hydrophobic pore-wall surface and sufficiently large platinum complex anions which behave like hydrophobic solutes. The behavior under the condition with the surface-induced phase transition occurrence (condition 1)

is substantially different from that under the condition without it (condition 2). Some of the important results are recapitulated below.

- (1) Under condition 2, the penetration process can be understood in terms of the concept of diffusion based on Fick's law in which the driving force is the concentration gradient of platinum complex anions. The penetration is slower when the pore diameter is smaller, the pore is deeper, and the anion size is larger.
- (2) Under condition 1, the penetration process can be interpreted only by applying statistical mechanics of confined molecular liquids. An effective surface-anion interaction, which is strongly attractive and long ranged, comes into play. Within the nanopore, it is even much stronger. As a physical quantity departs more from the transition point (e.g., the anion concentration increases and the pore diameter decreases), the effective interaction becomes further stronger. Anomalous behavior is exhibited as described in (3).
- (3) The penetration is faster when the pore diameter is smaller and the anion size is larger. The penetration rate remains unexpectedly fast even when the pore was much deeper. The penetration can be made faster using large coexisting cations with sufficiently high hydrophobicity.

We note that the behavior under condition 1 can be qualitatively opposite to that under condition 2.

The results obtained are of particular importance in the view of controlling the mass transfer within nanoporous media. In aqueous solution, the combination of a hydrophobic pore-wall surface and large metal ions with sufficiently high hydrophobicity is quite advantageous for drastically accelerating the ion penetration into the pores whose size is only a few nanometers. The acceleration can be assured irrespective of the pore depth. The use of large coexisting cations can lead to further acceleration. We believe that the present study sheds new light on the further development of such devices as batteries, electric double layer capacitors, and sensors. The physical insights highlighted should be applicable to general cases where chemical reactions within nanoporous materials play a pivotal role.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpcc.5b03852.

Further discussions on the phase transition in the bulk mixture and the surface-induced phase transition (PDF)

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Notes

The authors declare no competing financial interest.

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